



2012 Trial Examination

FORM VI

MATHEMATICS EXTENSION 2

Monday 6th August 2012

General Instructions

- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 100 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet and on the tear-off sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 85 boys

Examiner

KWM/BDD

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The value of $\int_0^{\pi} 5 \sin x \cos^4 x \, dx$ is:

- (A) 0 (B) 2 (C) -2 (D) 20

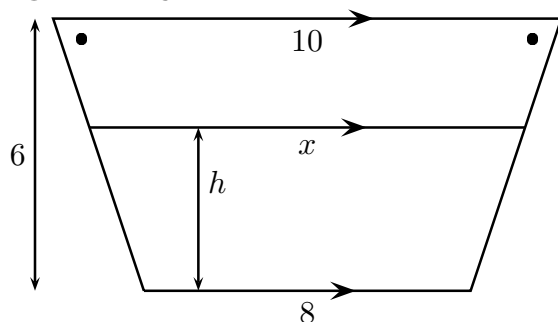
QUESTION TWO

The eccentricity of the ellipse $\frac{x^2}{100} + \frac{y^2}{36} = 1$ is $e = \frac{4}{5}$.

The distance between the two foci is:

- (A) 8 (B) 16 (C) 20 (D) 25

QUESTION THREE



The diagram above shows a trapezium with an interval x units in length drawn parallel to the base and h units from the base. An expression for x in terms of h is given by:

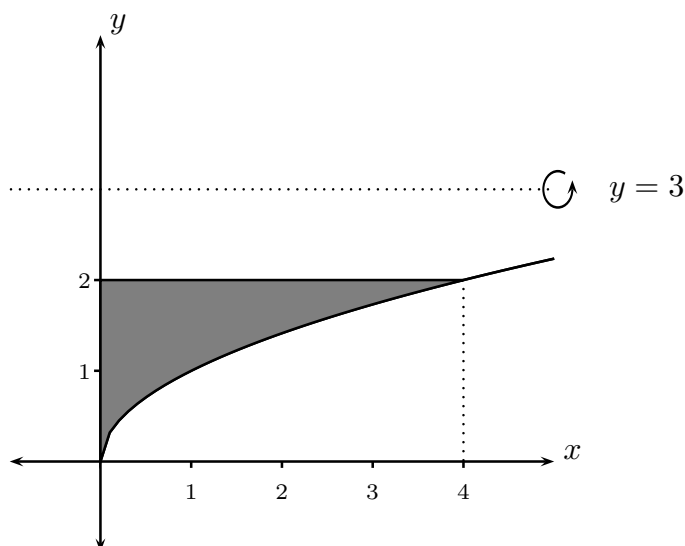
- (A) $x = 8 + \frac{h}{6}$
 (B) $x = 12 + \frac{h}{8}$
 (C) $x = 8 + \frac{h}{3}$
 (D) $x = 5 - \frac{6}{h}$

QUESTION FOUR

A motor bike and rider with mass 200 kg accelerates under a propelling force of 12 800 Newtons and as it moves it experiences a resisting force of $2v^2$ Newtons, where v m/s is the velocity. The motion is therefore described by the equation $\ddot{x} = 64 - \frac{v^2}{100}$.

What is the maximum speed attained by the bike?

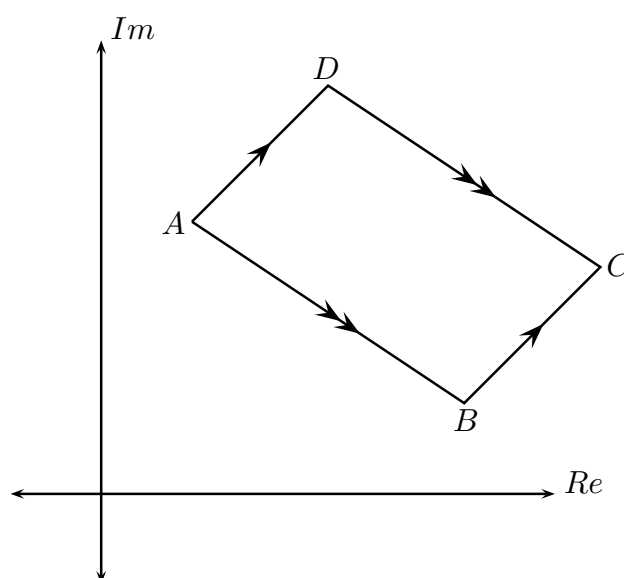
- (A) 288 km/h
- (B) 80 km/h
- (C) 280 km/h
- (D) $\sqrt{32}$ m/s

QUESTION FIVE

The area bounded by the curve $y = \sqrt{x}$, the y -axis and the line $y = 2$ is rotated about the line $y = 3$. The volume is to be calculated by taking slices perpendicular to the axis of rotation. Which integral gives the volume of the solid formed?

- (A) $\pi \int_0^2 \left((3 - y)^2 - 1 \right) dy$
- (B) $\pi \int_0^4 \left(x - 6\sqrt{x} + 8 \right) dx$
- (C) $\pi \int_0^4 \left(2 - 2\sqrt{x} + x \right) dx$
- (D) $\pi \int_0^4 \left((3 - x)^2 - 1 \right) dx$

QUESTION SIX



The diagram above shows parallelogram $ABCD$ drawn in the first quadrant of the complex plane. The points A , B and C represent the complex numbers z_1 , z_2 and z_3 respectively. The vector DB represents the complex number:

- (A) $z_1 + z_3 - 2z_2$
- (B) $z_2 - z_1 - z_3$
- (C) $2z_2 - z_1 - z_3$
- (D) $2z_2 - z_1 + z_3$

QUESTION SEVEN

The complex number ω is a root of the equation $z^3 + 1 = 0$. Which of the following is FALSE?

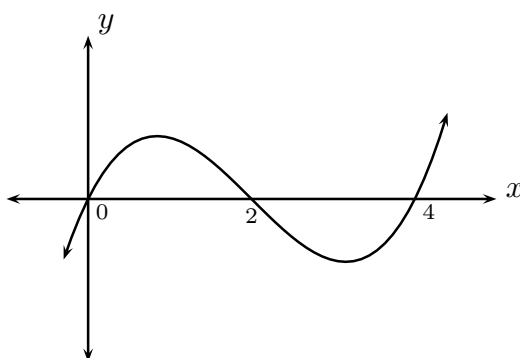
- (A) $\bar{\omega}$ is also a root.
- (B) $\omega^2 + 1 - \omega = 0$
- (C) $\frac{1}{\omega}$ is also a root.
- (D) $(\omega - 1)^3 = -1$

QUESTION EIGHT

The value of $\lim_{N \rightarrow \infty} \int_0^N e^{-x} dx$ is:

- (A) 0
- (B) 1
- (C) -1
- (D) ∞

QUESTION NINE



The graph of $y = f(x)$ is shown above. The graph of $y = f(2 - x)$ is:

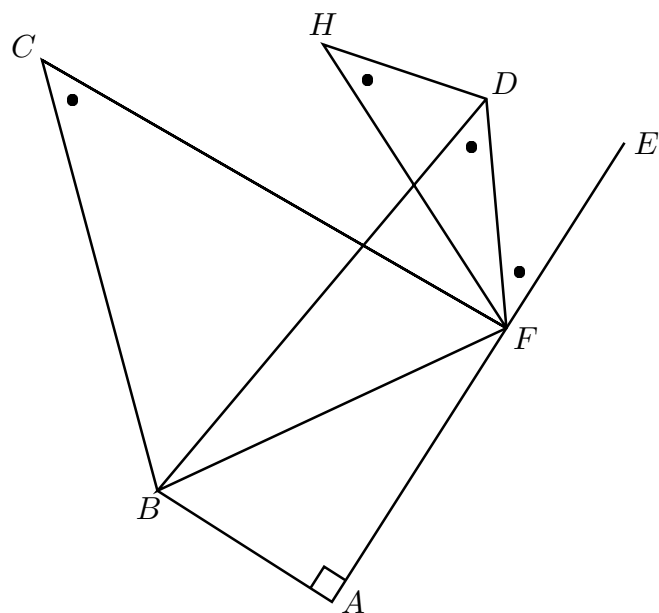
- (A)

(B)

(C)

(D)

QUESTION TEN



In the diagram above, all equal angles are marked with a dot and AFE is straight. Which of these statements is INCORRECT?

- (A) A circle may be drawn through A, B, F with diameter BF
- (B) The points B, C, D, F are concyclic.
- (C) A circle may be drawn through D, F, H with tangent AE
- (D) A circle may be drawn through B, D, F with tangent AE .

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

(a) Find $\int x \cos x \, dx$. 1

(b) Find $\int \frac{x+1}{x-2} \, dx$. 2

(c) Use the substitution $x = 2 \cos \theta$ to evaluate $\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{4-x^2}} \, dx$. 3

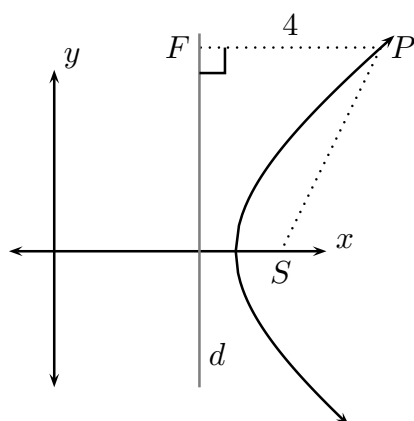
(d) (i) Find constants A , B and C such that 2

$$\frac{x^2 - x + 1}{(x+1)^2} = A + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}.$$

(ii) Hence find $\int \frac{x^2 - x + 1}{(x+1)^2} \, dx$. 2

(e) The polynomial $P(x) = 2x^3 - 3x^2 - 36x + k$ has a double zero. Find the possible values of k . 3

(f)



The diagram above shows the right branch of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$. The right directrix d and right focus S are shown. Let P be a fixed point on the curve and F a point on d , such that PF is perpendicular to d . It is known that $PF = 4$.

(i) Find the eccentricity of the hyperbola. 1

(ii) Find the distance PS . 1

Exam continues overleaf ...

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**(a) Let $z = 3 - i$ and $w = 2 + i$. Find the following in the form $x + iy$.

(i) $\overline{zw}$

1

(ii) $\left| \frac{z}{w} \right|$

1(b) (i) Find the two square roots of $16 - 30i$.**2**(ii) Hence solve $z^2 - 2z - (15 - 30i) = 0$.**1**

(c) In separate diagrams, sketch the region in the complex plane where:

(i) $|z - i| \leq |z - 1|$

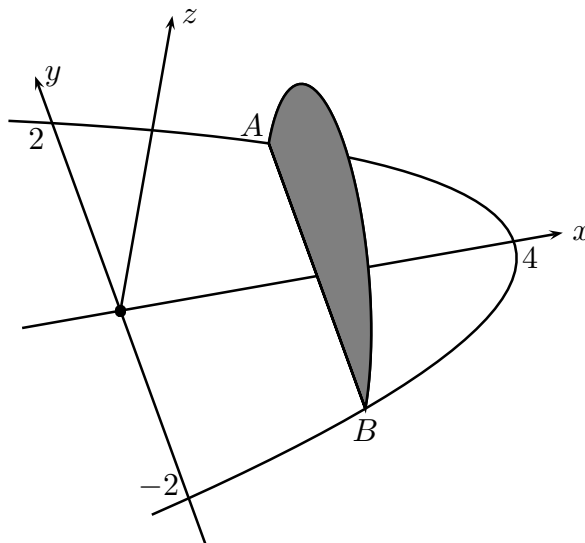
2

(ii) $0 < \arg(z - (1 + i)) \leq \frac{\pi}{3}$

2(d) (i) Write the complex number $1 + \sqrt{3}i$ in modulus-argument form.**1**(ii) Use de Moivre's theorem to express $(1 + \sqrt{3}i)^5$ in the form $a + bi$.**2**(e) The locus of a point P on the complex plane is defined by $|z - (1 + 2i)| = 3$.(i) Sketch the locus of P .**2**(ii) Find the maximum value of $|z|$.**1**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



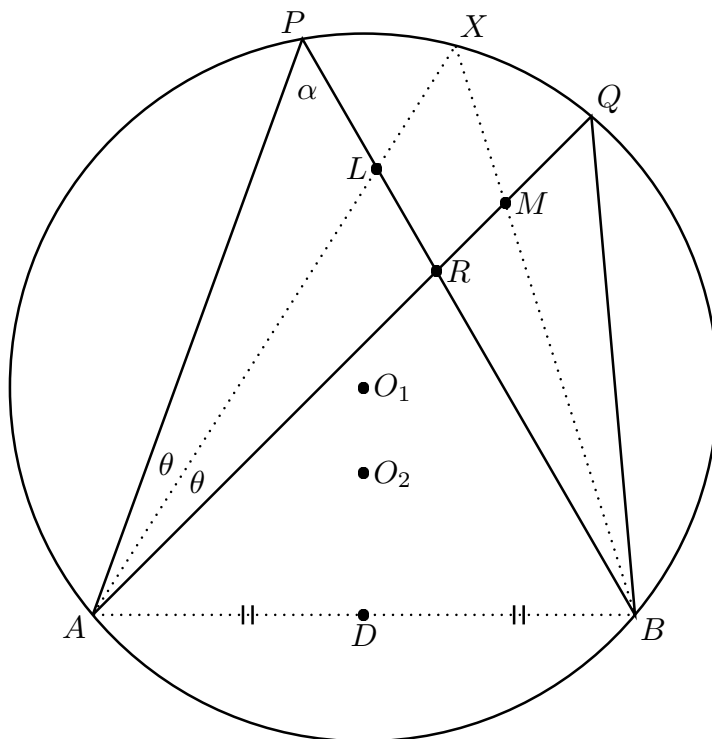
The base of a solid is the region bounded by the parabola $y^2 = 4 - x$ and the y -axis, as in the diagram above. A typical vertical cross-section is a semi-circle parallel to the y -axis.

- (i) Write an expression in terms of x for the area of a cross-section standing on the interval AB , as in the diagram. 1
- (ii) Find the volume of the solid. 2
- (b) Consider the polynomial equation $P(x) = 0$, where $P(x) = x^4 - 2x^3 + 6x^2 - 8x + 8$.
 - (i) Given that $x = 1 + i$ is a root of $P(x) = 0$, write down a second root. 1
 - (ii) Write $P(x)$ as a product of two quadratic expressions. 2
 - (iii) Fully factorise $P(x)$ over the complex numbers. 1
- (c) An object of mass 2 kilograms is projected vertically upwards from ground level at a speed of 20 m/s. It experiences a resistance of $\frac{v^2}{2}$ Newtons at a speed of v m/s, and reaches a maximum height H metres. Take upwards as positive and $g = 10 \text{ m/s}^2$.
 - (i) Show that the acceleration is given by $\ddot{x} = \frac{-40 - v^2}{4}$. 1
 - (ii) Show that the time it takes for the object to reach its maximum height is approximately 0.8 seconds. 2
 - (iii) Find the maximum height H reached by the object. 2
 - (iv) Calculate the speed of projection required to reach a maximum height of $2H$ metres. 3

Exam continues overleaf ...

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) (i) Find the roots of the equation $z^5 - 1 = 0$. You may leave the complex roots in modulus-argument form. 2
- (ii) Hence find the exact value of $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5}$. 2
- (b) The diagram below is reproduced on a tear off sheet at the end of this paper. It should be handed in with your solution to this question.



The minor arc AB on the circle $\mathcal{C}_1$ subtends angles at P and Q on $\mathcal{C}_1$. The point X is chosen on the minor arc PQ such that AX bisects $\angle PAQ$. Suppose that AX and BX intersect BP and AQ at L and M respectively, and that AQ and BP intersect at R . Let O_1 be the centre of $\mathcal{C}_1$, D be the midpoint of AB , $\angle PAX = \theta$ and $\angle APB = \alpha$.

- (i) Show that BX bisects $\angle PBQ$. 1
- (ii) Show that A , L , M and B lie on the circumference of a circle $\mathcal{C}_2$ and call its centre O_2 . 1
- (iii) By considering the perpendicular bisector of the chord AB , explain why O_1 , O_2 and D are collinear. 1
- (iv) Show that $\angle AO_2D = \alpha + \theta$. 1
- (v) Show that $\angle O_1AO_2 = \theta$. 1

QUESTION FOURTEEN (Continued)

- (c) Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$, and let $P(a \cos \theta, b \sin \theta)$ be a point on the ellipse in the first quadrant.

Let $O(0, 0)$ be the centre of the ellipse. Let T and N be the y -intercepts of the tangent and normal respectively to the ellipse at P . You may use the fact that the tangent at P has equation $ay \sin \theta + bx \cos \theta = ab$.

- (i) Show the normal to the ellipse at P has equation

2

$$by \cos \theta - ax \sin \theta = (b^2 - a^2) \cos \theta \sin \theta.$$

- (ii) Show that the product $TO \times ON$ is independent of θ .

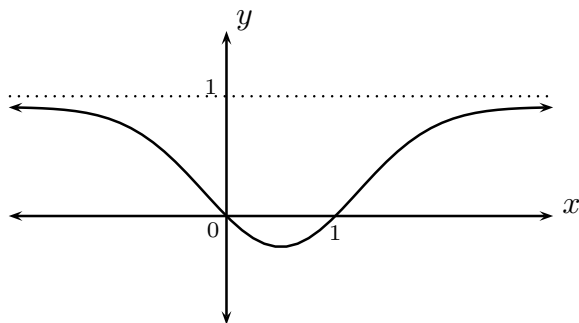
2

- (iii) Suppose that the circle with diameter TN is drawn. Using the result in (ii), or otherwise, show that the points of intersection of the circle with the x -axis are independent of θ .

2

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Consider the graph
- $y = f(x)$
- sketched below.



Draw separate one-third page sketches of the following graphs:

- (i) $y = \left(f(x)\right)^2$ 2
- (ii) $y = \ln f(x)$ 2
- (iii) $y = xf(x)$ 2

- (b) Let
- $I_n = \int_0^1 x^n \sqrt{1-x^3} dx$
- , for
- $n \geq 2$
- .

- (i) By writing
- $x^n \sqrt{1-x^3} = x^{n-2} \times x^2 \sqrt{1-x^3}$
- , or otherwise, show that
- 3

$$I_n = \frac{2n-4}{2n+5} \times I_{n-3}, \quad \text{for } n \geq 5$$

- (ii) Hence find
- I_8
- .
- 1

- (c) Let
- $z = \cos \theta + i \sin \theta$
- .

- (i) Show that
- $\sin n\theta = \frac{z^n - z^{-n}}{2i}$
- and
- $\cos n\theta = \frac{z^n + z^{-n}}{2}$
- .
- 2

- (ii) Hence, or otherwise, prove the identity
- 3

$$32 \sin^4 \theta \cos^2 \theta = \cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2.$$

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) The circle $x^2 + y^2 + ax + by + c = 0$ cuts the rectangular hyperbola $x = kt$, $y = k/t$ in four points P , Q , R and S defined by parameters p , q , r and s respectively.

- (i) Show that the parameters are roots of the quartic equation

1

$$k^2t^4 + akt^3 + ct^2 + bkt + k^2 = 0.$$

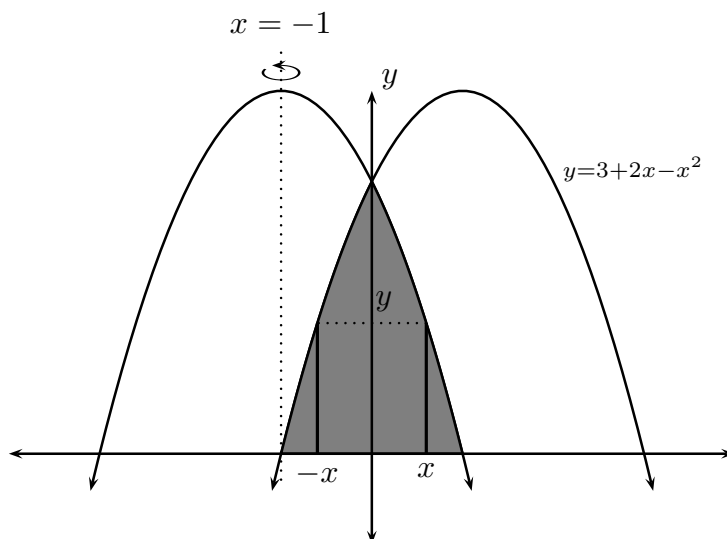
- (ii) Show that $pqrs = 1$.

1

- (iii) Show that if QS is a diameter of the hyperbola then PR is a diameter of the circle.

3

(b)



The diagram above shows the parabola $y = 3 + 2x - x^2$ and its reflection in the y -axis. The vertical strips shown will generate cylindrical shells of the same height y when rotated about the line $x = -1$.

- (i) Show that the sum of the areas of these two cylindrical shells is $4\pi y$.

1

- (ii) Find the volume of the solid formed when the region bounded by the parabolas and the x -axis is rotated about the line $x = -1$.

2

- (c) The polynomial equation $x^3 - px + 1 = 0$ has roots α , β and γ .

- (i) Find the cubic equation whose roots are $\frac{1}{\alpha^3}$, $\frac{1}{\beta^3}$ and $\frac{1}{\gamma^3}$.

2

- (ii) Find an expression for $\frac{\beta\gamma}{\alpha^5} + \frac{\alpha\gamma}{\beta^5} + \frac{\alpha\beta}{\gamma^5}$ in terms of p .

2

- (d) Use the identity $(1+x)^{2n+1}(1-x)^{2n} = (1+x)(1-x^2)^{2n}$ to prove that

3

$$\binom{2n+1}{0}\binom{2n}{0} - \binom{2n+1}{1}\binom{2n}{1} + \cdots + \binom{2n+1}{2n}\binom{2n}{2n} = (-1)^n \binom{2n}{n}.$$

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

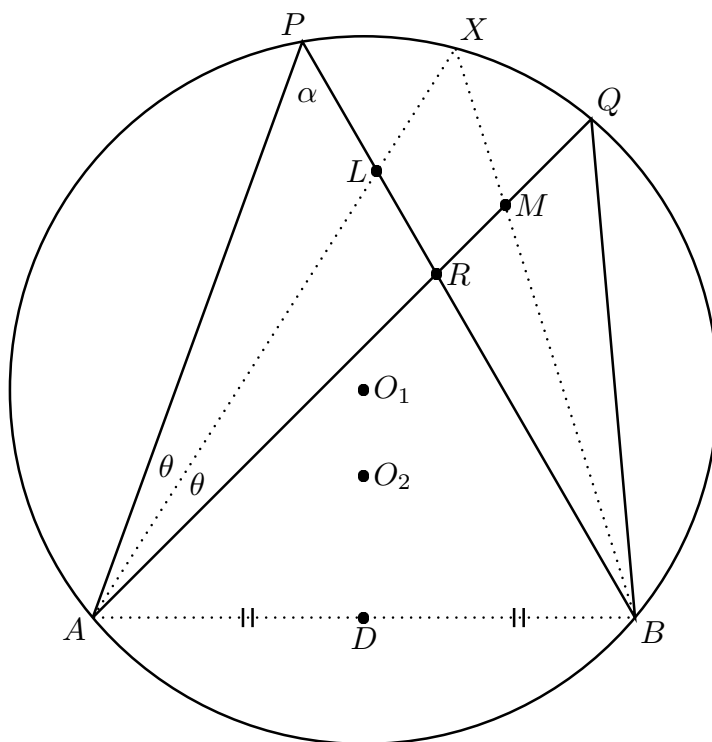
NOTE : $\ln x = \log_e x, \quad x > 0$

CANDIDATE NUMBER:

DETACH THIS SHEET AND BUNDLE IT WITH THE REST OF QUESTION FOURTEEN.

QUESTION FOURTEEN

- (b) The circle geometry diagram from question 14b is reproduced below. Hand this in with your solution to question 14b.



B L A N K P A G E



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

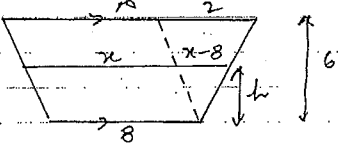
Question Ten

A ☐ B ☐ C ☐ D ☐

MULTIPLE CHOICE (1 mark each)

Q1/ $\int_0^\pi 5 \sin x \cos^4 x \, dx = - \left[\cos^5 x \right]_0^\pi$
 $= - \{ -1 - 1 \}$
 $= 2$ [B]

Q2/ $s'(-ae, 0)$ $s(ae, 0)$
 $d = 2ae$
 $d = \frac{2 \times 10 \times 4}{3}$
 $= 16$ [B]

Q3/ 
 $\frac{x-8}{2} = \frac{16}{6}$
 $x-8 = \frac{16}{3}$
 $x = 8 + \frac{16}{3}$ [C]

Q4/ $u = 64 - \frac{v^2}{100}$
 when $u = 0$, $v = 80 \text{ m/s}$
 $80 \text{ m/s} = 288 \text{ km/h}$ [A]

Q5/ $V = \pi \int_0^4 (3-y)^2 - 1^2 \, dy$
 $= \pi \int_0^4 8 - 6y + y^2 \, dy$ [B]

Q6/ $\vec{BC} = \vec{z}_3 - \vec{z}_2$
 $\vec{OD} = \vec{OA} + \vec{BC}$
 $= \vec{z}_1 + \vec{z}_3 - \vec{z}_2$
 Now $\vec{DB} = \vec{OB} - \vec{OD}$
 $= \vec{z}_2 - (\vec{z}_1 + \vec{z}_3 - \vec{z}_2)$
 $= 2\vec{z}_2 - \vec{z}_1 - \vec{z}_3$ [C]

Q7/ $z^3 + 1 = 0$
 roots: $w, \frac{1}{w}, -1$
 sum: $w + \frac{1}{w} - 1 = 0$
 $w^2 - w + 1 = 0$ [D]

Q8/ $\lim_{N \rightarrow \infty} \int_0^N e^{-x} \, dx$
 $= \lim_{N \rightarrow \infty} \left[-e^{-x} \right]_0^N$
 $= \lim_{N \rightarrow \infty} \left(-\frac{1}{e^N} \right) - -1$
 $= 1$ [B]

Q9/ $y = f(x)$
 $y = f(2-x)$
 Reflect about the line $x=1$ [A]

Q10/ The angle between the chord and tangent at the point of contact F is NOT equal to the angle in the alternate segment. ($\angle FBD \neq \angle DFE$) [D]

(10)

QUESTION 11
 (a) $\int x \cos x \, dx = x \sin x - \int \sin x \, dx$
 $= x \sin x + \cos x + c$
 b) $\int \frac{x+1}{x-2} \, dx = \int \frac{x-2+3}{x-2} \, dx$
 $= \int 1 + \frac{3}{x-2} \, dx$
 $= x + 3 \ln|x-2| + c$

c) $\int \frac{1}{x^2 \sqrt{4-x^2}} \, dx$
 let $x = 2 \cos \theta$
 $dx = -2 \sin \theta \, d\theta$
 when $x = 1$, $\theta = \frac{\pi}{3}$
 when $x = \sqrt{3}$, $\theta = \frac{\pi}{6}$
 $\int_{\frac{\pi}{3}}^{\frac{\pi}{6}} \frac{-2 \sin \theta \, d\theta}{4 \cos^2 \theta \sqrt{4-4 \cos^2 \theta}}$
 $= \int_{\frac{\pi}{3}}^{\frac{\pi}{6}} \frac{-2 \sin \theta \, d\theta}{8 \cos^2 \theta \sin \theta}$
 $= -\frac{1}{4} \int_{\frac{\pi}{3}}^{\frac{\pi}{6}} \sec^2 \theta \, d\theta$
 $= -\frac{1}{4} \left[\tan \theta \right]_{\frac{\pi}{3}}^{\frac{\pi}{6}}$
 $= -\frac{1}{4} \left\{ \tan \frac{\pi}{6} - \tan \frac{\pi}{3} \right\}$
 $= -\frac{1}{4} \left\{ \frac{1}{\sqrt{3}} - \sqrt{3} \right\}$
 $= -\frac{1}{4} \left(\frac{\sqrt{3}}{3} - \sqrt{3} \right) = \frac{\sqrt{3}}{6}$

(d)(i) $x^2 - x + 1 \equiv A(x+1)^2 + B(x+1) + C$
 put $x = -1$: $C = 3$
 put $x = 0$: $A + B + 3 = 1$
 $A + B = -2$
 put $x = 1$: $4A + 2B + 3 = 1$
 $2A + B = -1$
 ② - ①: $A = 1$, $\therefore B = -3$

(ii) The integral becomes:
 $\int 1 - \frac{3}{x+1} + \frac{3}{(x+1)^2} \, dx$
 $= x - 3 \ln|x+1| - \frac{3}{x+1} + c$

(e) $P(x) = 2x^3 - 3x^2 - 36x + k$
 $P'(x) = 6x^2 - 6x - 36$
 $x^2 - x - 6 = 0$
 $(x-3)(x+2) = 0$
 $P(3) = P(-2) = 0$
 $P(3): 54 - 27 - 108 + k = 0$
 $k = 81$
 or
 $P(-2): -16 - 12 + 72 + k = 0$
 $k = -44$

(f)(i) $b^2 = a^2(e^2 - 1)$
 $9 = 16(e^2 - 1)$
 $\frac{9}{16} = e^2 - 1$
 $\frac{25}{16} = e^2$
 $e = \frac{5}{4}$ (15)

(ii) $\frac{PS}{PF} = e$
 $\frac{PS}{4} = \frac{5}{4}$, $\therefore PS = 5$ with

QUESTION 12

$$(a) z = 3-i \quad w = 2+i$$

$$(i) \overline{zw} = \overline{(3-i)(2+i)} \\ = \overline{7+i} \\ = 7-i \checkmark$$

$$(ii) \left| \frac{z}{w} \right| = \frac{|z|}{|w|} = \frac{\sqrt{10}}{\sqrt{5}} \\ = \sqrt{2} \checkmark$$

$$(b) \text{ let } z = a+ib \\ (a+ib)^2 = 1-30i$$

$$R: a^2 - b^2 = 1$$

$$Im: 2ab = -30$$

$$a=5, b=3 \quad \text{or}$$

$$a=-5, b=3$$

$$z = 5-3i \quad \text{or} \quad z = -5+3i \checkmark$$

$$(ii) z^2 - 2z - (15-30i) = 0$$

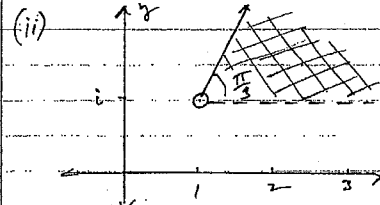
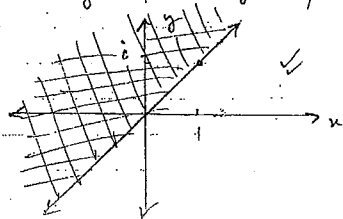
$$\Delta = b^2 - 4ac \\ = 4 + 4(15-30i) \\ = 4(1+15-30i) \\ = 4(16-30i) \\ = 2^2(5-3i)^2 \quad (i)$$

$$z = \frac{-2 \pm 2(5-3i)}{2}$$

$$z = 1 \pm (5-3i) \quad \checkmark$$

$$z = 6-3i \quad \text{or} \quad z = -4+3i$$

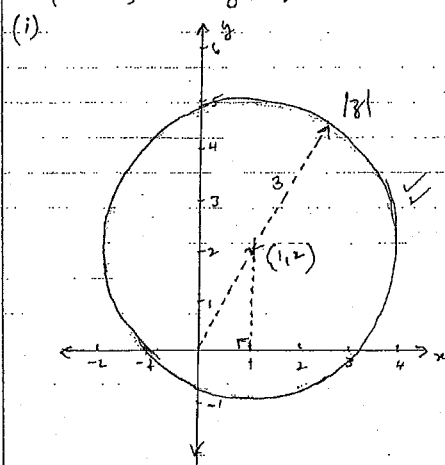
$$(c) (i) |z-i| \leq |z-1|$$



$$(d) (i) 1 + \sqrt{3}i = 2 \cos \frac{\pi}{3} \checkmark$$

$$(ii) (1 + \sqrt{3}i)^5 = (2 \cos \frac{\pi}{3})^5 \\ = 32 \cos \frac{5\pi}{3} \checkmark \\ = 32 \left(\frac{1}{2} - \frac{\sqrt{3}i}{2} \right) \\ = 16(1 - \sqrt{3}i) \checkmark$$

$$(e) |z - (1+2i)| = 3 \\ (x-1)^2 + (y-2)^2 = 9$$



$$(ii) |z|_{\max} = 3 + \sqrt{5} \checkmark$$

(15)

QUESTION 13

$$(a) (i) A = \frac{\pi}{2} y^2$$

$$A = \frac{\pi}{2} (4-x) \checkmark$$

$$(ii) V = \frac{\pi}{2} \int_0^4 (4-x) dx \\ V = \frac{\pi}{2} \left[4x - \frac{x^2}{2} \right]_0^4 \checkmark$$

$$V = 4\pi \text{ cubic units} \checkmark$$

$$(b) x^4 - 2x^3 + 6x^2 - 8x + 8 = 0$$

$$(i) 1-i \text{ is another root.} \checkmark$$

$$(ii) x^2 - 5x + 4 = 0 \\ x^2 - 2x + 2 \text{ is a quadratic factor.}$$

$$\begin{array}{r} x^2 - 2x + 2 \overline{) x^4 - 2x^3 + 6x^2 - 8x + 8} \\ \underline{x^4 - 2x^3 + 2x^2} \\ 4x^2 - 8x + 8 \end{array} \checkmark$$

$$P(x) = (x^2 - 2x + 2)(x^2 + 4) \checkmark$$

$$(iii) P(x) = (x-1-i)(x-1-i)x(x-2i)(x+2i) \checkmark$$

$$(c) (i) m\ddot{x} = -mg - \frac{v^2}{2}$$

$$2\ddot{x} = -20 - \frac{v^2}{2} \checkmark$$

$$\ddot{x} = -10 - \frac{v^2}{4}$$

$$(ii) \frac{dv}{dt} = -\frac{v^2 + 40}{4}$$

$$\frac{dv}{v^2 + 40} = -\frac{1}{4} dt$$

$$t = -4 \tan^{-1} \frac{v}{\sqrt{40}} + C \checkmark$$

$$\text{when } t=0, v=20$$

$$C = \frac{2}{\sqrt{10}} \tan^{-1} \sqrt{10}$$

$$t = \frac{2}{\sqrt{10}} \tan^{-1} \sqrt{10} - \frac{2}{\sqrt{10}} \tan^{-1} \frac{v}{\sqrt{10}} \checkmark$$

$$\text{max height when } v=0:$$

$$t = \frac{2}{\sqrt{10}} \tan^{-1} \sqrt{10}$$

$$t \approx 0.8 \text{ s}$$

$$(iii) \ddot{u} = -\frac{40 - v^2}{4}$$

$$v \frac{dv}{du} = -\frac{v^2 + 40}{4}$$

$$\frac{dv}{v} = -\frac{v^2 + 40}{4v^2}$$

$$\frac{dv}{v} = -\frac{40}{v^2} \checkmark$$

$$u = -2 \ln(v^2 + 40) + C$$

$$\text{when } u=0, v=20$$

$$C = 2 \ln(440)$$

$$u = 2 \ln \left(\frac{440}{v^2 + 40} \right)$$

$$\text{max height when } v=0$$

$$u = 2 \ln 11 \text{ m.} \checkmark \checkmark$$

$$(iv) u = -2 \ln(v^2 + 40) + C$$

$$\text{from (i). when } u=0, \text{ put } v=V$$

$$u = 2 \ln \left(\frac{V^2 + 40}{v^2 + 40} \right) \checkmark$$

$$\text{view height} = 24 \\ = 4 \ln \text{ m}$$

$$2 \ln \left(\frac{v^2 + 40}{v^2 + 40} \right) = 4 \ln 11 \quad \checkmark$$

$$\frac{v^2 + 40}{v^2 + 40} = 121$$

put $v=0$ for max.
height.

$$v^2 + 40 = 40 \times 121$$

$$v^2 = 120 \times 40$$

$$v^2 = 4800$$

$$v = 40\sqrt{3} \text{ m/s} \quad \checkmark$$

(15)

QUESTION 14

(a) (i) $z^5 = 1$

let $z = \text{cis } \theta$

$(\text{cis } \theta)^5 = 1$

Using de Moivre's thm:

$\text{cis } 5\theta = 1$

Re: $\cos 5\theta = 1$

$5\theta = 2\pi n \quad \checkmark$

$\theta = \frac{2\pi n}{5}, n=0, \pm 1, \pm 2$

Roots are:

$1, \text{cis}\left(\frac{2\pi}{5}\right), \text{cis}\left(-\frac{2\pi}{5}\right), \text{cis}\left(\frac{4\pi}{5}\right)$
and $\text{cis}\left(-\frac{4\pi}{5}\right) \quad \checkmark$

(ii) $\text{cis}\left(\frac{2\pi}{5}\right) + \text{cis}\left(-\frac{2\pi}{5}\right) = 2\cos\frac{2\pi}{5}$

$\text{cis}\left(\frac{4\pi}{5}\right) + \text{cis}\left(-\frac{4\pi}{5}\right) = 2\cos\frac{4\pi}{5}$

Sum of roots $\checkmark = -\frac{b}{a} = 0$

$2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} + 1 = 0$

$\therefore \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2} \quad \checkmark$

(b) (i) $\angle PAX = \angle PBX = \theta$

$\angle XAQ = \angle XBQ = \theta$

(Angles drawn to the circumference standing on the same arc are equal.) $\angle PBX = \angle XBQ$

$\therefore BX$ bisects $\angle PBQ. \quad \checkmark$

(ii) $\angle LAR = \theta$ (given)

$\angle LMB = \theta$ (i)

Angles in the alternate segment are equal, hence $\angle LMB$ are conjugate points and lie on the circumference of $C_2. \quad \checkmark$

(iii) Since AB is a common chord of circles C_1 and C_2 , the perpendicular bisector of chord AB passes through the centres O_1 and O_2 . Hence O_1, O_2 and D are collinear. $\checkmark$

(iv) $\angle ALR = \theta + \alpha$
(exterior angle of triangle thm.)

$\angle AO_2B = 2(\theta + \alpha)$

(angle at the centre is twice the angle drawn to the circumference of C_2 standing on the arc AB .)

$\triangle AO_2D \equiv \triangle BO_2D$ (SSS)

$\angle AO_2D = \angle BO_2D \quad \checkmark$

(corresponding angles of cong. triangles)

$\angle AO_2D + \angle BO_2D = 2(\theta + \alpha)$

$2\angle AO_2D = 2(\theta + \alpha)$

$\therefore \angle AO_2D = \theta + \alpha$

(v) $\angle AO_1B = 2\alpha$ (angle drawn to the centre is twice the angle drawn to the circumference.)

$\angle AO_1D = \angle BO_1D$

(matching angles of cong. triangles)

$\angle AO_1D = \alpha$, but

$\angle AO_2D = \alpha + \theta$ (part iv)

$\therefore \angle O_1AO_2 = \theta$ (external angle of triangle theorem. $\checkmark$)

... cont.

Q14...

6.

(e)

$$(i) \quad x = a \cos \theta \quad y = b \sin \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta \quad \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= -\frac{b}{a} \frac{\cos \theta}{\sin \theta}$$

gradient of the normal at $P(a \cos \theta, b \sin \theta)$ is

$$m = \frac{a \sin \theta}{b \cos \theta} \quad \text{Using } \checkmark$$

$$y - y_1 = m(x - x_1)$$

$$y - b \sin \theta \cos \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

by $\cos \theta - b^2 \sin \theta \cos \theta = a x \sin \theta - a^2 \sin \theta \cos \theta$ Using the intercepts of chords.
 by $\cos \theta - a x \sin \theta = (b^2 - a^2) \sin \theta \cos \theta$ theorem:

$$RO \times OS = TO \times ON$$

$$s^2 = a^2 - b^2$$

$$s = \sqrt{a^2 - b^2}$$

hence:

$$R(-\sqrt{a^2 - b^2}, 0) \text{ and } S(\sqrt{a^2 - b^2}, 0)$$

both co-ordinates are independent of θ .

$$(ii) \quad a y \sin \theta + b x \cos \theta = ab$$

To find T put $x=0$.

$$y = \frac{b}{\sin \theta}$$

$$T(0, \frac{b}{\sin \theta})$$

$$\text{by } \cos \theta - a x \sin \theta = (b^2 - a^2) \sin \theta \cos \theta$$

To find N put $x=0$.

$$N(0, \frac{b^2 - a^2}{b} \sin \theta)$$

$$TO = \frac{b}{\sin \theta}$$

$$ON = -\frac{(b^2 - a^2)}{b} \sin \theta \quad (\text{y co-ordinate of } N \text{ is negative.})$$

$$TO \times ON = \frac{b}{\sin \theta} \times \frac{(a^2 - b^2)}{b} \sin \theta$$

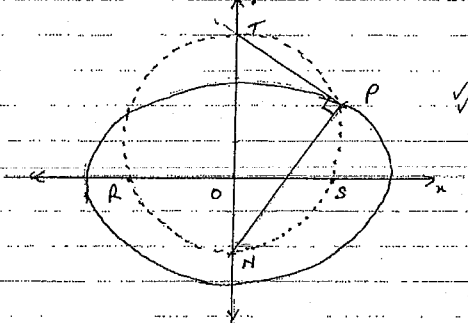
$$= a^2 - b^2 \quad \checkmark$$

(independent of θ).

(iii) $\angle TPN = 90^\circ$, hence

TN is a diameter of the circle. (angle in a semi circle.)

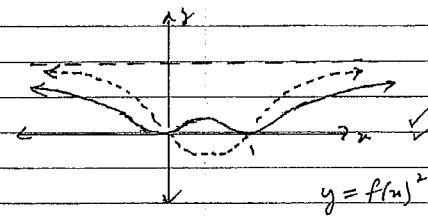
Let $S(s, 0)$ and $R(-s, 0)$ be the x -intercepts of the circle.
 $OS = OR = s$



QUESTION 15

8

(a)



$$I_n = \int_0^1 u^{n-2} x u^2 \sqrt{1-u^2} du$$

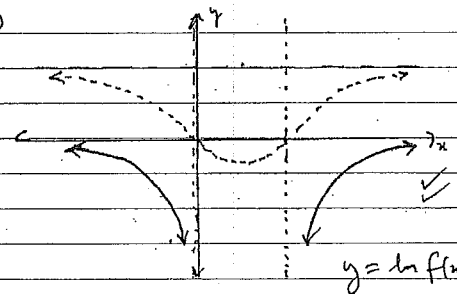
$$\text{let } u = x^{n-2}$$

$$u' = (n-2)x^{n-3}$$

$$v' = u^2 (1-u^2)^{1/2}$$

$$v = -\frac{2}{9} (1-u^2)^{3/2}$$

(b)



$$I_n = \left[uv \right]_0^1 + \frac{2}{9} (n-2) \int_0^1 u^{n-3} (1-u^2)^{3/2} du$$

$$I_n = \frac{2}{9} (n-2) \int_0^1 u^{n-3} (1-u^2)^{3/2} du \quad \checkmark$$

$$I_n = \frac{2}{9} (n-2) \int_0^1 u^{n-3} \sqrt{1-u^2} (1-u^2) du$$

$$I_n = \frac{2}{9} (n-2) \left\{ \int_0^1 u^{n-3} \sqrt{1-u^2} du - \int_0^1 u^n \sqrt{1-u^2} du \right\}$$

$$I_n = \frac{2}{9} (n-2) \{ I_{n-3} - I_n \} \quad \checkmark$$

$$I_n \left\{ 1 + \frac{2}{9} (n-2) \right\} = \frac{2}{9} (n-2) I_{n-3}$$

$$I_n (9 + 2n - 4) = 2(n-2) I_{n-3}$$

$$I_n = \frac{2n-4}{2n+5} I_{n-3} \quad \checkmark$$

$$(ii) I_2 = \int_0^1 u^2 \sqrt{1-u^2} du$$

$$= -\frac{2}{9} \left[(1-u^2)^{3/2} \right]_0^1$$

$$= \frac{2}{9}$$

(b)

$$(i) I_n = \int_0^1 u^n \sqrt{1-u^2} du$$

$n \geq 2$

$$I_5 = \frac{10-4}{10+5} \cdot I_2 = \frac{6}{15} \times \frac{2}{9}$$

$$I_8 = \frac{16-4}{16+5} \cdot I_5 = \frac{12}{21} \times \frac{6}{15} \times \frac{2}{9}$$

$$= \frac{16}{315} \quad \checkmark$$

(c)

$$(1) \quad z = \cos \theta + i \sin \theta$$

$$(2) \quad z^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$(3) \quad z^{-n} = \cos n\theta - i \sin n\theta$$

$$(1) + (2) \quad z^n + z^{-n} = 2 \cos n\theta$$

$$\cos n\theta = \frac{z^n + z^{-n}}{2}$$

$$(1) - (2): \quad z^n - z^{-n} = 2i \sin n\theta$$

$$\sin n\theta = \frac{z^n - z^{-n}}{2i}$$

(ii)

$$\text{LHS} = 32 \sin^4 \theta \cos^2 \theta$$

$$= 32 \left(\frac{z - z^{-1}}{2i} \right)^4 \left(\frac{z + z^{-1}}{2} \right)^2$$

$$= \frac{1}{2} (z - \frac{1}{z})^4 (z + \frac{1}{z})^2$$

$$= \frac{1}{2} \left[(z - \frac{1}{z})(z + \frac{1}{z}) \right]^2 (z - \frac{1}{z})^2$$

$$= \frac{1}{2} (z^2 - \frac{1}{z^2})^2 (z^2 - 2 + \frac{1}{z^2})$$

$$= \frac{1}{2} \left\{ (z^4 - 2 + \frac{1}{z^4}) (z^4 - 2 + \frac{1}{z^4}) \right\}$$

$$= \frac{1}{2} \left\{ z^8 - 2z^4 + z^2 - 2z^2 + 4 - \frac{2}{z^2} + \frac{1}{z^4} - \frac{2}{z^4} + \frac{1}{z^6} \right\}$$

$$= \frac{1}{2} \left\{ (z^6 + \frac{1}{z^6}) - 2(z^4 + \frac{1}{z^4}) - (z^2 + \frac{1}{z^2}) + 4 \right\}$$

$$= \frac{1}{2} \left\{ 2 \cos 6\theta - 4 \cos 4\theta - 2 \cos 2\theta + 4 \right\}$$

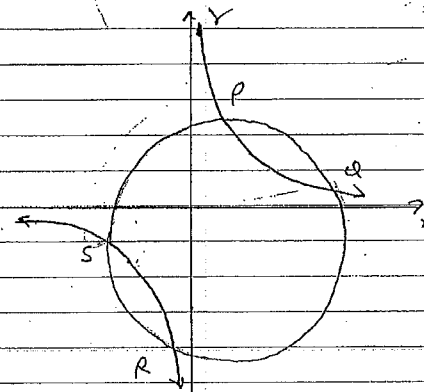
$$= \cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2$$

= RHS

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QUESTION 16

a)



$$(1) \quad x = kt \quad \text{and} \quad y = \frac{k}{t}$$

$$x^2 + y^2 + ax + by + c = 0$$

$$\therefore k^2 t^2 + \frac{k^2}{t^2} + akt + \frac{bk}{t} + c = 0$$

$$k^2 t^4 + k^2 + akt^3 + bkt + ct^2 = 0$$

$$k^2 t^4 + akt^3 + ct^2 + bkt + k^2 = 0$$

all points P, Q, R, S satisfy this equation.

$$(ii) \quad \text{Product of the roots} = \frac{c}{a}$$

$$PQ \cdot RS = \frac{k^2}{k^2}$$

$$PQ \cdot RS = 1$$

(iii) If RS is a diameter of the hyperbola,

$$Q(\frac{k}{t}, \frac{t}{k}) \quad S(s, \frac{s}{t})$$

$$\text{Hence: } qt = -st \quad \text{and}$$

$$\frac{q}{t} = -\frac{s}{t}$$

$$\text{Hence: } q = -s$$

Now gradient of P(KP, K)

$$S(KS, \frac{K}{S})$$

$$M_{PS} = \frac{\frac{K}{S} - KP}{KS - KP}$$

$$= \frac{P - S}{S(P - P)} \quad \checkmark$$

$$SP(S - P)$$

$$= -\frac{1}{SP}$$

$$\text{Similarly } M_{QS} = -\frac{1}{PS}$$

$$M_{PS} \times M_{QS} = \frac{1}{PS \cdot SP} \quad \text{Since } (S = -P)$$

$$\text{product} = \frac{1}{PS \cdot SP} = \frac{1}{PS \cdot (-P)} = -\frac{1}{PS^2}$$

$$= -\frac{1}{PS^2} \quad \checkmark \quad (PS \cdot SP = 1)$$

$\therefore \angle PSR = 90^\circ$ and PR is a diameter.

(b)(i) (angle in a semi-circle.)

$$A_T = 2\pi r y + 2\pi r y$$

$$R = n+1 \quad \text{and } r = 1 - \pi$$

$$A_T = 2\pi y \left\{ n+1 + 1 - \pi \right\}$$

$$A_T = 2\pi y \times 2$$

$$A_T = 4\pi y \quad \text{(as required)}$$

$$(ii) \quad V = 4\pi \int_0^1 (3 - 2x - x^2) dx$$

$$V = 4\pi \left[3x - x^2 - \frac{x^3}{3} \right]_0^1$$

$$V = 4\pi \left[\left(3 - 1 - \frac{1}{3} \right) - (0) \right]$$

$$V = \frac{20\pi}{3} \quad \text{cubic units. } \checkmark$$

Q16...

11

$$(i) x^3 - px + 1 = 0$$

$$(1) \left(\frac{1}{x^{1/3}}\right)^3 - p\left(\frac{1}{x^{1/3}}\right) + 1 = 0 \quad \checkmark$$

$$\frac{1}{x} + 1 = \frac{p}{x^{1/3}}$$

$$\left(\frac{1}{x} + 1\right)^3 = \frac{p^3}{x}$$

$$\frac{1}{x^3} + \frac{3}{x^2} + \frac{3}{x} + 1 = \frac{p^3}{x}$$

$$1 + 3x + 3x^2 + x^3 = \frac{p^3}{x^4}$$

$$x^3 + (3-p)x^2 + 3x + 1 = 0 \quad \checkmark$$

$$(ii) x^3 - px + 1 = 0$$

$$\alpha\beta\gamma = -1$$

$$\frac{\beta\gamma}{\alpha^5} + \frac{\alpha\gamma}{\beta^5} + \frac{\alpha\beta}{\gamma^5}$$

$$= \alpha\beta\gamma \left\{ \frac{1}{\alpha^6} + \frac{1}{\beta^6} + \frac{1}{\gamma^6} \right\}$$

$$= \alpha\beta\gamma \left\{ \left(\frac{1}{\alpha^3} + \frac{1}{\beta^3} + \frac{1}{\gamma^3} \right)^2 - \right.$$

$$\left. 2 \left\{ \frac{1}{\alpha^3\beta^3} + \frac{1}{\alpha^3\gamma^3} + \frac{1}{\beta^3\gamma^3} \right\} \right\}$$

$$= - \left\{ (p^3 - 3)^2 - 2 \times 3 \right\}$$

$$6 - (p^6 - 6p^3 + 9)$$

$$6p^3 - p^6 - 3 \quad \checkmark$$

P.T.O.

Q16 (d) →

12

$$LHS = (1+x)^{2n+1} (1-x)^{2n}$$

$$= \sum \left\{ \binom{2n+1}{0} + \binom{2n+1}{1}x + \dots + \binom{2n+1}{2n-1}x^{2n-1} + \binom{2n+1}{2n}x^{2n} + \binom{2n+1}{2n+1}x^{2n+1} \right\}$$

$$\times \left\{ \binom{2n}{0} - \binom{2n}{1}x + \binom{2n}{2}x^2 - \dots + \binom{2n}{2n-2}x^{2n-2} - \binom{2n}{2n-1}x^{2n-1} + \binom{2n}{2n}x^{2n} \right\}$$

Consider the coefficient of x^{2n} in the product:

$$= \binom{2n+1}{0} \binom{2n}{2n} - \binom{2n+1}{1} \binom{2n}{2n-1} + \dots + \binom{2n+1}{2n} \binom{2n}{0}$$

$$\text{Using } \binom{n}{r} = \binom{n}{n-r} \quad \checkmark$$

$$\text{Coefficient of } x^{2n} = \binom{2n+1}{0} \binom{2n}{0} - \binom{2n+1}{1} \binom{2n}{1} + \dots + \binom{2n+1}{2n} \binom{2n}{2n}$$

$$\text{Now RHS} = (1+x)(1-x^2)^{2n}$$

$$= (1+x) \left\{ \binom{2n}{0} - \binom{2n}{1}x^2 + \binom{2n}{2}x^4 - \dots + (-1)^n \binom{2n}{n}x^{2n} + \dots + \binom{2n}{2n}x^{4n} \right\}$$

No term in x^{2n} will be generated by the product of x in $(1+x)$ and the expansion since terms in the expansion have even powers. $\checkmark$

$$\text{Coefficient of } x^{2n}: \quad (-1)^n \binom{2n}{n}$$

Hence

$$\binom{2n+1}{0} \binom{2n}{0} - \binom{2n+1}{1} \binom{2n}{1} + \dots + \binom{2n+1}{2n} \binom{2n}{2n} = (-1)^n \binom{2n}{n}$$

(15)